A Quartic in Ninety-Eight Variables under the Rational *abc* Conjecture

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Abstract

Assuming the rational *abc* conjecture and the specified elliptic-group, pole-realization and simultaneous prime-pattern inputs, we construct a family of integer quartics in ninety-eight total variables. Every sufficiently large fixed odd input shift gives rational input projection exactly $\mathbb{Z}$. The quartic is a sum of eighty-six primitive quadratic squares and has exactly 882 monomials. Its coefficient height, for a shift $B \geq 3$, is $\max(11600064, 2B^4 + 169997)$. Six rational contacts symmetric about one permit a short quadratic evaluation and normalization. A height estimate for a sum of sixth powers bounds the input together with one elliptic index. All six local tests have order one; only plain elliptic formal parameters are used. The sufficient shift is ineffective.

1 Statement and arithmetic assumptions

For a reduced fraction u/v, with $v > 0$, put $h(u/v) = \log \max(|u|, v)$. Write $h(1 : u : v)$ for the logarithmic projective height of the indicated rational point. The coefficient height of an integer polynomial is the maximum absolute value of its coefficients. All variable counts include the distinguished input.

Theorem 1.1. *Under the assumptions below, there is a computable family*

$$P_B \in \mathbb{Z}[X, Y_1, \ldots, Y_{97}], \qquad B \geq 3 \text{ odd},$$

such that every member has degree exactly four, content one, exactly 882 *monomials, and coefficient height*

$$H(P_B) = \max(11600064, 2B^4 + 169997).$$

There is a constant B_* such that, for every odd $B \geq B_*$,

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{97}, \quad P_B(x, \boldsymbol{y}) = 0.$$

The integer B is a fixed coefficient parameter. No numerical sufficient value of B_* is determined. In particular, the coefficient fixture $B = 3$ is not asserted to define the integers.

Let $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, let σ be conjugation, and put

$$d = 3 - \rho, \quad D = d^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = (x_P, y_P) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

The arithmetic assumptions have the following precise scope.

1. The rational *abc* conjecture: for every $\epsilon > 0$ there is $K_\epsilon \geq 1$ such that $c \leq K_\epsilon \operatorname{rad}(abc)^{1+\epsilon}$ for pairwise coprime positive integers $a + b = c$.

2. The elliptic source gives $2E(K) = \mathbb{Z}P$, with P of infinite order, and the index-height comparison stated in [2]. The square-abscissa characterization proved in [4] then applies to all nonzero multiples of P.

3. For a nonzero rational b, set $B_0 = 17b$. The pole equations

$$\begin{aligned} p_0^2 &= B_0(1-e_0)(1+3e_0), \\ B_0 q_0^2 &= (s_0^2 - B_0 e_0(B_0 - b^2))(s_0^2 + 3B_0 e_0(B_0 - b^2)) \end{aligned} \tag{1}$$

exclude negative odd valuations of b outside $\{2, 3, 17\}$. For inputs satisfying

$$\begin{gathered} 0 < b < 17, \qquad b \text{ has square reduced denominator}, \\ b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2, \end{gathered} \tag{2}$$

the retained restricted-converse and descent hypotheses give a witness with $p_0 q_0 s_0 \neq 0$. The hypotheses and their nonzero-witness consequence are specified in [2, 7, 8].

4. The simultaneous prime-pattern input has exactly the quantified form in [2, 3]: arbitrary fixed rational contact coefficients after denominator clearing, the prescribed Gaussian congruences and conjugate reduction orders, fixed avoidance and divisibility, and compatible real approximation. Its analytic source is Sun's specified Theorem 4.1 and the estimates pinned in those papers. All scales and finite preparation data are fixed before prime selection.

The exponent-nineteen contact-height input is not used here.

The one-point case of the proved aggregate trace-height theorem [5], obtained from von Buhren's family height theorem, gives constants $A > 0$, $J \geq 0$ such that

$$n^2 \leq A h(T(n)) + J, \qquad T(n) = \operatorname{Tr} x([n]P), \quad n \in \mathbb{Z} \setminus \{0\}. \tag{3}$$

The additive constant is ineffective. Each abscissa is a positive square at both real embeddings and exceeds d there, so $T(n) > 6$.

2 Two height consequences of *abc*

Set

$$\Theta_6 = \{1 \pm 1, 1 \pm 2, 1 \pm 3\} = \{-2, -1, 0, 2, 3, 4\}.$$

For $t \notin \Theta_6$, let $M(t)$ be the product of primes $p > 76176$ for which $v_p(t - \theta) > 0$ is odd for some $\theta \in \Theta_6$.

Lemma 2.1. *Assume rational abc. There is a fixed $C_0 \geq 0$ such that*

$$h(t) \leq \frac{2}{3} \log M(t) + C_0.$$

Proof. For a primitive pair (u, v), $v > 0$, use the squarefree binary form $G(u, v) = \prod_{\theta \in \Theta_6} (u - \theta v)$ of degree six. Granville's consequence of *abc* [1, Proposition 1], with $\epsilon = 1/4$, gives

$$\log \operatorname{rad}(G(u, v)) \geq \frac{15}{4} \log \max(|u|, v) - C.$$

For a nonzero integer m, let $\operatorname{sf}(m)$ be the product of primes having odd valuation in m. Prime by prime, $\operatorname{rad}(m)^2 \leq |m| \operatorname{sf}(m)$. Also $|G(u, v)| \leq C' \max(|u|, v)^6$. Outside the fixed cutoff, no prime divides two distinct factors of G, and a prime dividing v divides none of them. Its

squarefree part outside the cutoff is therefore precisely $M(u/v)$. The omitted squarefree factors are bounded by a fixed product of primes. Consequently

$$\log M(u/v) \geq \left(2 \cdot \frac{15}{4} - 6\right) h(u/v) - C'' = \frac{3}{2} h(u/v) - C'',$$

as required. □

Lemma 2.2. *Assume rational abc. There is a fixed $C_1 \geq 0$ such that*

$$h(1 : u : v) \leq \frac{2}{5} h(u^6 + v^6) + C_1 \qquad (u, v \in \mathbb{Q}).$$

One may take $C_1 = \frac{7}{30} \log 2 + \frac{11}{30} \log K_{1/11}$.

Proof. The zero-coordinate cases follow from $h(w^6) = 6h(w)$. Otherwise write $u = gA/D_0$, $v = gB/D_0$, where $\gcd(A, B) = \gcd(g, D_0) = 1$, $g, D_0 > 0$. Put $H = \max(|A|, |B|)$ and $N = A^6 + B^6$, so $H^6 \leq N \leq 2H^6$. Let $k = \gcd(N, D_0^6)$, $m_0 = N/k$, $n_0 = D_0^6/k$. Then $u^6 + v^6 = g^6 m_0/n_0$ is reduced. Write $M = \max(g^6 m_0, n_0)$.

Every prime dividing N either divides D_0 or survives in m_0, so $\mathrm{rad}(N) \leq D_0 m_0$. Apply *abc* with $\epsilon = 1/11$ to $A^6 + B^6 = N$. The three positive terms are pairwise coprime, and

$$H^{7/2} \leq K_{1/11}^{11/12} D_0 m_0.$$

Using $D_0^6 = n_0 N/m_0 \leq 2n_0 H^6/m_0$ gives

$$H^{5/2} \leq 2^{1/6} K_{1/11}^{11/12} m_0^{5/6} n_0^{1/6}.$$

Moreover $\max(gH, D_0) \leq 2^{1/6} H (M/m_0)^{1/6}$. Taking logarithms and combining yields

$$\begin{aligned} h(1 : u : v) &= \log \max(gH, D_0) \\ &\leq C_1 + \frac{1}{6} \log m_0 + \frac{1}{15} \log n_0 + \frac{1}{6} \log M \leq C_1 + \frac{2}{5} \log M. \end{aligned}$$

This includes common numerator factors and all denominator cancellation. □

3 Quadratic point equations

Put $\beta = 2X^2 + B$. If $X = a/b$ is reduced, integrality of β implies $b^2 \mid 2$, hence $b = 1$. Thus $\beta \in \mathbb{Z}$ if and only if $X \in \mathbb{Z}$; for integer input β is odd, and always $\beta \geq B$.

All equations in this section are over K and mean their two rational component equations. Products of stored quantities have degree at most two. For the root point retain $h, j, a \in K$ and impose

$$a = h^2, \qquad j^2 = a^2 - D. \tag{4}$$

Then $A = (a, hj) = [n]P$ for some nonzero integer n by the square-abscissa theorem [4]. Retain $\chi, f, S \in K$ with

$$\chi f = 1, \qquad S^2 = \chi - Df. \tag{5}$$

They give $H = (\chi, \chi S) \in E(K)$. Write $H = [m]R_0 + T_0$, $T_0 \in E[2](K)$, and put $h_* = m/2$. This is an index notation; it does not mean multiplication of a point by a noninteger rational number.

3.1 A pair of points with fixed difference

Retain $V, Q, W_A, I_A, K_A \in K$ satisfying

$$\begin{aligned} hI_A &= (j + DW_A)V + aQ, \\ hK_A &= -(j + DW_A)Q - aV, \\ VQ &= aW_A, \\ V^2 + Q^2 + 2jW_A + DW_A^2 &= 1. \end{aligned} \tag{6}$$

Lemma 3.1. *Equations (4) and (6) describe points $U, V_1 \in \mathbb{Z}P$ with $U - V_1 = A$, allowing either numerator point to be the identity. Both I_A and K_A are nonzero. On the nonidentity branches,*

$$U = \left(\frac{a}{V^2}, \frac{aI_A}{V^3}\right), \qquad V_1 = \left(\frac{a}{Q^2}, \frac{aK_A}{Q^3}\right), \qquad z(U) = -\frac{V}{I_A}, \quad z(V_1) = -\frac{Q}{K_A},$$

where $z = -x/y$. In all branches the quantities $R_U, R_1 \in K$ defined by

$$I_A^2 = a - DaR_U, \qquad K_A^2 = a - DaR_1 \tag{7}$$

are $f(U)^2, f(V_1)^2$, with $f = 1/x$ extended by zero at the identity.

Proof. The square chart gives $haj \neq 0$. Squaring the first two equations and using the last two gives $aI_A^2 = a^2 - DV^4$ and $aK_A^2 = a^2 - DQ^4$. For $VQ \neq 0$ these are exactly the two curve equations. Their abscissas are squares, so the points lie in $\mathbb{Z}P$. The line with slope $(1/W_A - j)/h$ through U also passes through $-V_1$ and $-A$. Substitution gives its three abscissas $a/V^2, a/Q^2, a$, counted with multiplicity. The chord-and-tangent law therefore gives $U - V_1 = A$, including the tangent case. Nontorsion of the nonidentity points implies $I_A K_A \neq 0$.

If $V = 0$, then $W_A = 0$, $Q^2 = 1$, $I_A = hQ$ and $K_A = -jQ/h$. The interpretation is $U = O, V_1 = -A$. If $Q = 0$, then $W_A = 0$, $V^2 = 1$, $I_A = jV/h$ and $K_A = -hV$, giving $U = A, V_1 = O$. The two zero cases cannot occur together. The asserted nonvanishing and reciprocal-square identities also hold there.

Conversely, let $U, V_1 \in \mathbb{Z}P$ be nonidentity points with $U - V_1 = A$. Their abscissas are squares. The line through $U, -V_1, -A$ has equation $y = \mu x + c$, with $c = -h(\mu h + j)$. Put $R = \mu h + j$. Vieta's identities give $x(U)x(V_1) = R^2$ and $x(U) + x(V_1) = \mu^2 - a$; in particular $R \neq 0$. Choose square roots $u_0^2 = x(U)$, $v_0^2 = x(V_1)$ with $u_0 v_0 = R$, and put

$$V = h/u_0, \qquad Q = h/v_0, \qquad W_A = 1/R, \qquad I_A = y(U)V^3/a, \qquad K_A = y(V_1)Q^3/a.$$

The two Vieta identities give the last two chart equations, and the line ordinates give the first two. This also covers tangent lines. The identity branches are supplied by the explicit formulas above. □

3.2 Subtraction of the fixed point

Retain $g, F_0, z_1 \in K$ and impose

$$\begin{aligned} 8(x_P\chi + D)g + (39 - 36\rho) + (8 - 14\rho)S &= 0, \\ fF_0 = g^2, \qquad K_A z_1 + Q &= 0. \end{aligned} \tag{8}$$

On the admissible domain, $F_0 = f(H - P)$ and $z_1 = z(V_1)$. Define the affine expressions

$$\begin{aligned} N &= (-312 - 69\rho) + (-236 - 120\rho)S, \\ D_0 &= (-576 - 256\rho) + (312 + 69\rho)S + (504 - 1344\rho)g. \end{aligned} \tag{9}$$

Retain $L, L_0 \in K$ with

$$I_A L = VS, \qquad L_0 N = z_1 D_0. \tag{10}$$

After the exclusions proved below, these mean

$$L = \frac{z(U)}{z(H)}, \qquad L' = \sigma L_0 = \sigma\left(\frac{z(V_1)}{z(H-P)}\right).$$

No coordinate for $z(H-P)$ is retained.

For clarity, the fixed subtraction formulas follow from the usual addition slope $(\chi S + y_P)/(\chi - x_P)$. Direct reduction using $S^2 = \chi - D/\chi$ gives

$$f(H-P) = \chi g^2, \qquad z(H-P) = N/D_0.$$

Before multiplication by the common unit $3 + 2\rho$, the numerator is $32(y_P - x_P S)$ and the denominator is $(-704 + 384\rho) + (660 - 417\rho)S + (6888 - 5040\rho)g$. The numerator vanishes only when $S = y_P/x_P$, and the curve equation then gives

$$(\chi - x_P)(\chi + D/x_P) = 0.$$

The latter alternative includes $H = -P + (0,0)$ and must be excluded, along with $H = P$, before division. An exact function-field check of these identities is included with the coefficient certificate.

4 Six tests and two comparison scales

Set

$$C = L - L', \qquad I = L - \beta, \qquad T = \operatorname{Tr} a, \qquad a_0 = 1, \quad a_1 = \beta^6 + T^6.$$

The six field tests are $C, I, f(H), f(U), \sigma f(H-P), \sigma f(V_1)$. Retain the four reciprocal squares, C^2, I^2, and the fourth powers $f(U)^4, \sigma f(V_1)^4$ as quadratic outputs. The two fields in (7) supply two reciprocal squares directly. Four scalar outputs compute the comparison scale without taking any root or making any division:

$$s_h = \beta^2 + T^2, \quad p_h = \beta T, \quad v_h = s_h^2 - 3p_h^2, \quad a_1 = s_h v_h = \beta^6 + T^6.$$

Retain a rational λ. For $a \in \{a_0, a_1\}$ put $t = a\lambda$, $\delta = t - 1$, and use scalar coordinates r, q, c, k satisfying

$$r = \delta^2, \qquad c = qr, \qquad k = cr, \qquad kr - 14k + 49c - 36q = r^2. \tag{11}$$

Define, without storing another coordinate,

$$F = (r-1)(r-4)(r-9) = r^3 - 14r^2 + 49r - 36 = \prod_{\theta \in \Theta_6} (t - \theta).$$

The displayed quadratic equations imply exactly $Fq = r^2$ and $k = qr^2$. They compute the needed normalization without a coordinate for F. Store q^2, q^4 and the two field factors

$$\begin{aligned} T_1 &= 1 + qC^2 + q^2 f(H)^2 + q^4 f(U)^4, \\ T_2 &= 1 + qI^2 + q^2 \sigma f(H-P)^2 + q^4 \sigma f(V_1)^4, \qquad W = T_1 T_2 = w_0 + w_1 \rho. \end{aligned} \tag{12}$$

Finally retain scalars τ, b with

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad kb = \tau^2. \tag{13}$$

The normalized pole uses seven further rational coordinates e, u, p, v, w, p_2, w_2 and five quadratic equations

$$\begin{aligned} p_2 = p^2, \qquad 17bp_2 = 289 + 34e - 3e^2, \\ e(17-b) = 17uv, \qquad w_2 = w^2, \qquad 17bw_2 = u^2 + 34uv - 867v^2. \end{aligned} \tag{14}$$

For $b \neq 0$ these give (1) under

$$e_0 = e/17, \quad p_0 = bp, \quad s_0 = bu, \quad q_0 = b^2uw.$$

Conversely a witness with $s_0 \neq 0$ gives the displayed normalized coordinates by these formulas and $v = e(17-b)/(17u)$. This explains both the pole obstruction and its use in completeness.

Lemma 4.1. *Every rational solution of the quadratic source has $\delta F q k \neq 0$ in both detectors. Consequently*

$$q = \frac{\delta^4}{F}, \qquad k = \frac{\delta^8}{F} = \frac{1}{R}, \qquad R = \frac{F}{\delta^8}.$$

Proof. If $F = 0$, then $r^2 = \delta^4 = 0$, whereas at $t = 1$ the contact product equals $-36 \neq 0$. If $\delta = 0$, the equations give $q = 0$, $T_1 = T_2 = W = 1$, $\tau = 1/2$, and $k = 0$, contradicting $kb = \tau^2$. The remaining assertions follow directly. In particular $\lambda \neq 0$ because zero is a contact. □

5 Local extraction and denominator exclusions

At a prime $p > 76176$, the curve has good reduction and D is a unit. For a nonzero finite reciprocal abscissa $f(Q)$, its valuation at either place of K above p is even. If $v(x) < 0$, the curve equation gives $2v(y) = 3v(x)$; if $v(x) > 0$, it gives $2v(y) = v(x)$. The unit case is immediate. Zero reciprocal abscissas are interpreted as having infinite valuation. Before the exceptional H classes are excluded, the coordinate F_0 need not yet be interpreted as $f(H-P)$. Its even valuation nevertheless follows directly from $fF_0 = g^2$ and the even valuation of $f = f(H)$; $F_0 = 0$ is allowed. Thus the valuation argument below applies to the literal equations before any division in the fixed subtraction chart.

Let $p \mid M(t)$, put $\nu = v_p(R)$, and normalize valuations by $v(p) = 1$. Then ν is positive and odd, δ is a unit, and $v(q) = -\nu$. For each factor in (12), the three nonconstant finite valuations belong respectively to

$$1 + 2\mathbb{Z}, \qquad 2 + 4\mathbb{Z}, \qquad 4 + 8\mathbb{Z}.$$

Thus all term valuations, including zero for the constant term, are distinct. Each factor is nonzero and has valuation at most zero; valuation zero implies residue one. A zero test merely removes a term.

Proposition 5.1. *At every contact prime, there is one place at which all six tests have positive valuation. The four reciprocal-abscissa tests have valuation at least two.*

Proof. The preceding argument gives $W \neq 0$. Equation (13) then implies $w_0 \neq 0$, since $w_0 = 0$ would imply $w_1 = 0$. Hence

$$\tau = \frac{\mathrm{N}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0, \qquad b = R\tau^2 \neq 0.$$

Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality gives $v_p(b) \leq \nu + 2\ell$. This valuation is odd, so the pole obstruction forces $0 < v_p(b) \leq \nu + 2\ell$, hence $\ell > -\nu/2$. At a place attaining ℓ, each

factor valuation is at least ℓ. Each of its terms therefore has valuation greater than $-\nu/2$. For the three columns this gives respectively

$$v(C), v(I) > \nu/4, \qquad v(f(H)), v(\sigma f(H-P)) > 3\nu/4, \qquad v(f(U)), v(\sigma f(V_1)) > 7\nu/8.$$

Integer rounding and evenness give the assertion. □

The finite H chart may contain torsion translates. Before interpreting the subtraction formulas or either ratio, exclude

$$E[2](K) \ \cup \ (P + E[2](K)) \ \cup \ (-P + E[2](K)).$$

Every permitted point in this finite set has reciprocal abscissa a unit above $p > 76176$. The torsion translation formulas, together with

$$\mathrm{N}(x_P) = 41^2/64, \quad \mathrm{N}(x_P - d) = 17^2/64, \quad \mathrm{N}(x_P + d) = 63^2/64,$$

give this directly. If H belonged to the set, Proposition 5.1 would force both contact radicals to equal one. Lemmas 2.1 and 2.2, and $h(a_1) \le h(a_1\lambda) + h(\lambda)$, would then give $h(1 : \beta : T) \le \frac{4}{5}C_0 + C_1$. Since $\beta \ge B$, this is impossible for every sufficiently large fixed B.

Consequently $h_* \notin \{-1, 0, 1\}$ and all required subtraction denominators are nonzero. In particular $x_P\chi + D \neq 0$, $N \neq 0$, $D_0 \neq 0$, and $z(H), z(H-P) \neq 0$. The formulas above have their stated point interpretation. This exclusion does not use a logarithmic ratio and therefore precedes, rather than assumes, its validity.

6 Soundness for all sufficiently large shifts

Write $U = [u]P$, so $V_1 = [u-n]P$. At a contact place supplied by Proposition 5.1, H, U lie in the primary formal group and $H - P, V_1$ in the conjugate formal group. The formal parameter satisfies

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) \equiv 1 \pmod{p^{4v(z(Q))}}. \tag{15}$$

This is the constant-term case of the proved formal-logarithm estimate in [6]. The unit is defined to be one at the identity.

The integral group identities

$$[m]U = [2u]H, \qquad [m-2]V_1 = [2(u-n)](H-P)$$

kill the torsion component of H. Their full logarithmic ratios are $r = u/h_*$ and $r' = (u-n)/(h_*-1)$. At least one of h_*, h_*-1 is a p-adic unit, so at least one full ratio is integral. The units in (15) preserve the corresponding approximate valuation, and the positive-order test $C = L - L'$ transfers integrality to the other approximate ratio and hence to both full ratios. Only after this step does (15) give absolute congruences $L \equiv r$ and $L' \equiv r' \pmod{p}$. Therefore

$$n = h_*r - (h_*-1)r' \equiv L \equiv \beta \pmod{p}.$$

The argument also covers identity numerator points and half-integral h_*, since p is odd. In particular β is integral at p.

If $\beta \neq n$, let N_β be the positive absolute reduced numerator of $\beta - n$. Both $M(\lambda)$ and $M(a_1\lambda)$ divide N_β. Lemma 2.1 gives

$$h(a_1) \le h(\lambda) + h(a_1\lambda) \le \frac{4}{3}\log N_\beta + 2C_0.$$

Put $H_2 = h(1 : \beta : T)$. Lemma 2.2 implies

$$H_2 \le \frac{8}{15}\log N_\beta + C_2, \qquad C_2 = \frac{4}{5}C_0 + C_1.$$

For a reduced fraction $\beta = a/b$, $|a - nb| \le (1 + |n|)\max(|a|, b)$. Since $h(\beta) \le H_2$,

$$\log N_\beta \le H_2 + \log(1 + |n|), \qquad H_2 \le \frac{8}{7}\log(1 + |n|) + \frac{15}{7}C_2.$$

The trace-height bound (3) now yields

$$n^2 \le \frac{8A}{7}\log(1 + |n|) + \frac{15A}{7}C_2 + J.$$

Its right side grows logarithmically, so $|n|$ and then H_2 are uniformly bounded. Every constant here is independent of B. Since $\beta \ge B$, choosing one sufficiently large odd shift excludes $\beta \ne n$. Thus $\beta = n \in \mathbb{Z}$, and consequently $X \in \mathbb{Z}$. Together with the earlier exceptional-point exclusion this proves soundness for every odd B beyond one fixed threshold.

7 Simultaneous completeness

Fix $X \in \mathbb{Z}$ and an odd $B \ge 3$. Put $n = \beta$ and $A = [n]P$; this fixes T and both positive rational scales. Choose a nonzero reference integer divisible by 64 and compact real and finite-place neighborhoods for the points

$$H = [h]P, \quad U = [hn]P, \quad H - P = [h - 1]P, \quad V_1 = [(h - 1)n]P$$

which avoid the finitely many chart denominators. In these neighborhoods every rational test has a fixed lower valuation bound at the finitely many preparation primes. All these choices precede prime selection.

Choose a fixed $D_1 > 0$ clearing $D_1^2\theta/a$ for both scales and all contacts, and sufficiently divisible at the preparation primes. Apply the retained prime-pattern input to the distinct coefficients $-D_1^2\theta/a$, with repetitions removed, and write

$$\lambda = \frac{u}{D_1^2 w^2}, \qquad L_1 = D_1^2 w^2.$$

Its prime values are $u - (\theta/a)L_1$. The zero contact makes u one of the selected primes. The supplied conjugate reduction-order conditions and the compatible congruence progression in [2] give h for which H, U are formal at one place above each selected prime and $H - P, V_1$ are formal at the conjugate place. The fixed congruence and real neighborhoods are retained. Only natural depth one is requested at either branch.

Both full logarithmic ratios equal n. Hence the four reciprocal tests have valuation at least two and C, I have valuation at least four at the chosen place. Since $v_p(q) = -1$, every nonconstant term of both factors in (12) has positive valuation there. At the other place each factor is either a unit of residue one or has negative valuation. Thus $\mathrm{Tr}(W^{-1})$ has residue one or two and is a unit. The pole input has valuation one at every selected contact prime.

For arbitrary positive rational a the exact identity

$$R = \frac{a^6 \prod_{\theta \in \Theta_6}(u - (\theta/a)L_1)}{L_1^6(a\lambda - 1)^8} \tag{16}$$

has only square factors apart from the selected prime numerators. In particular no square condition on a is needed. At every other prime $v_p(R)$ is even, and the reciprocal trace changes valuations by an even number. These facts give a square reduced denominator for b.

At a preparation prime, increasing the fixed denominator divisibility makes $v_p(t) \ll 0$. Then

$$v_p(q) = v_p(R) = -2v_p(t), \qquad R = \delta^{-2} \prod_{\theta \in \Theta_6} \left(1 + \frac{1-\theta}{\delta}\right).$$

The product is arbitrarily close to one and is a local square at $2, 3, 17$. All weighted test terms are small, so $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two. Sufficient divisibility therefore gives the three local squareclasses and valuation depths in (2).

On the fixed compact real neighborhoods, as the lower bound on λ tends to infinity, $R \sim q \sim t^{-2}$, $W \to 1$, and $b \sim 1/(4t^2)$. Both real embeddings of every factor are positive for sufficiently large λ. Hence $0 < b < 17$ and every required denominator is nonzero. These bounds are uniform on the chosen compact sets, so they can be fixed before applying the prime-pattern theorem. Each prepared b has a nonzero pole witness, which converts to (14). All quadratic coordinates can now be filled by their stated rational values. This proves completeness for every integer input and every odd $B \geq 3$.

8 The polynomial and its exact certificate

Each field value represents two rational coordinates. For the abscissas a, χ, the four reciprocal squares and C^2, the stored second coordinate is one half of the ρ coefficient. Other stored field values use their ordinary coefficients. These rational changes of coordinates are invertible; they are fixed in the coefficient tables. Each component equation is cleared to a primitive integer quadratic, with positive first nonzero coefficient in the prescribed monomial order. Their sum of squares defines P_B.

The coordinate count is

Coordinates	Number
Input, its square, and the common slope	3
Root chart for A	6
Chart for H	6
Fixed subtraction quantities g, F_0	4
Pair chart for U, V_1	10
$z(V_1)$ and the two ratios	6
Four reciprocal squares and two comparison squares	12
Two fourth powers	4
Four scalar comparison-scale outputs	4
Two detector blocks, including their pole coordinates	42
The second argument $a_1\lambda$	1
Total	98

Each detector uses two contact outputs r, c, three weight coordinates, four factor components, two product components, the three scalars τ, k, b, and seven pole coordinates: twenty-one in all. There are 86 primitive quadratic equations, so over $\mathbb{Q}$ their sum of squares vanishes exactly when they all vanish.

The complete specification is in `computation/quartic/family_abc98`. The builder and independent checker reconstruct all 86 rows and every coefficient of the full quartic. There are 885 terms when powers of B are stored separately, and 882 ordinary monomials for every positive integer B. The coefficient of X^4 is one, proving degree four and content one. Only three ordinary coefficients contain more than one power of B: the constant coefficient $2B^4 + 169997$, the coefficient $48B^2 + 1$ of the square of the input-square coordinate, and $12B^2 - 19$ at the square of the rational component of the root abscissa. All remaining nonconstant coefficient

profiles are single powers of B; their maximal multipliers at powers one, two and three are respectively $64, 48, 16$. The largest parameter-free coefficient has absolute value 11600064. It follows directly that

$$H(P_B) = \max(11600064, 2B^4 + 169997) = 2B^4 + 169997 \quad \text{for odd } B \geq 49.$$

No coefficient cancellation changes the stated support for positive integer B.

The independent checker imports no producer. It also reconstructs the fixed-subtraction compression and both normalizations. A separate PARI computation verifies the fixed subtraction identities in the exact function field $K(\chi)[S]/(S^2 - \chi + D/\chi)$ against the elliptic addition law. These are finite algebraic certificates. The arithmetic assumptions and the ineffective sufficient shift remain those stated in Section 1.

References

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[4] *Square abscissas and direct charts for the elliptic subgroup*, companion paper, `papers/square_abscissa`.

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[8] *Nonzero witnesses in the pole equations*, companion paper, `papers/pole_parameters/pole_witness_density_20261109_v14.tex`.